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BERT WEIGHT: Word-Level Sentiment Analysis through Contextual Token Attribution and Embedding Weighting

Introduction

With the massive growth of online reviews and digital text, understanding sentiment at a granular level has become increasingly valuable. Existing tools like VADER and AFINN are limited by static lexicons, while BERT-based models, though powerful, remain a black box - relying on a single [CLS] token and offering little word-level transparency. Our project, BERT WEIGHT, tackles the challenge of sentiment analysis by ensuring both accuracy and explainability at the individual word level. We go deeper than traditional sentence-level analysis and focus on how words shape the sentiment of the given text.

Contributions

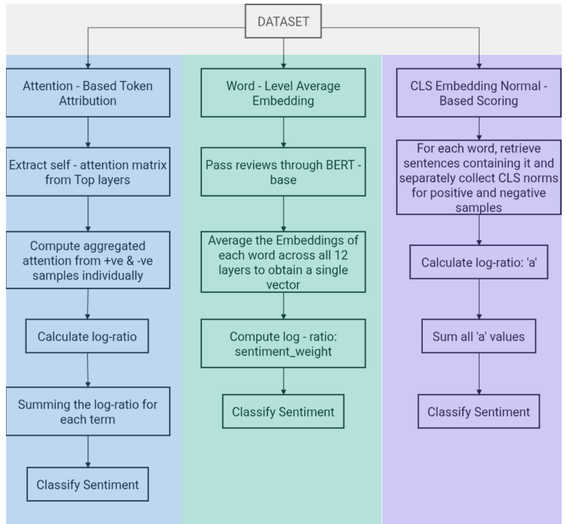
Three approaches were developed to assign sentiment scores to individual words using BERT:

- Attention-Based Attribution - extracts self-attention scores from BERT's top layers to identify which words drive positive or negative sentiment
- Word-Level Average Embedding - averages a word's embeddings across all 12 BERT layers and compares them across positive and negative reviews
- CLS Norm Scoring - uses the L2 norm of the [CLS] token from sentences containing a specific word to compute its sentiment weight

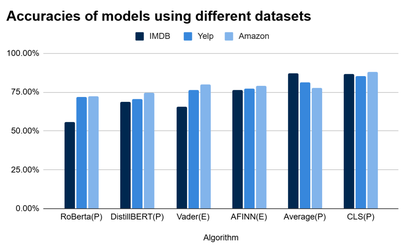
A log-based formula taken from SentiTFIDF to calculate the word scores was used. Below is the generalized version

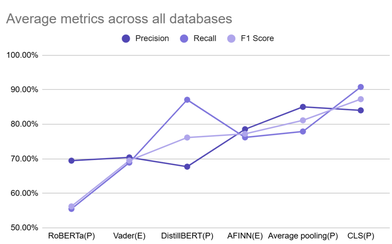
$$wordWeight = log_2\left(\frac{score_{pos} + e}{score_{neg} + e}\right)$$

Architecture



Results





We tested our proposed models on three datasets - Amazon, IMDB, Yelp- using 10-fold cross-validation.

- CLS Norm Scoring method achieved 86.77% accuracy which is the highest.
- An existing method VADER despite having high accuracy frequently misclassified neutral words as negative.
- Even in other metrics, namely Precision, Recall and F1 Score, the CLS Norm Scoring method outperforms the rest.
- A variant of BERT, the RoBERTa model performs the worst out of all, as it is built for fine-tuning. and uses a different tokenization method.

Conclusion

BERT WEIGHT represents a significant step forward in making sentiment analysis both powerful and interpretable. By moving away from opaque sentence-level classifications, our framework assigns meaningful sentiment scores to individual words - giving clear visibility into exactly which terms drive a positive or negative prediction. The proposed models outperformed traditional models like VADER and AFINN. The modular design makes BERT WEIGHT adaptable to real-world applications like product review analysis, and customer feedback monitoring among many more.

Github